



Original Article

Extending OpenMC validation to spent fuel canisters: A criticality benchmark against MCNP

J. Ruiz-Pineda^a, J. Romero-Barrientos^{a,b}, F. Molina^{a,b,c}, M. Zambra^a,
F. López-Usquiano^{a,b,c}

^a Centro de Investigación en Física Nuclear y Espectroscopía de Neutrones CEFNEN, Comisión Chilena de Energía Nuclear, Nueva Bilbao 12501, Las Condes, Santiago, Chile

^b Millennium Institute for Subatomic physics at high energy frontier - SAPHIR, Fernández Concha 700, Las Condes, Santiago, Chile

^c Departamento de Ciencias Físicas, Universidad Andres Bello, Sazié 2212, 837-0136, Santiago, Chile

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ABSTRACT

OpenMC is an open-source Monte Carlo code with increasing relevance in criticality safety and reactor physics applications. While its validation has covered a broad range of systems, its performance in spent nuclear fuel storage scenarios remains limited in the literature. This work benchmarks OpenMC against MCNP for eleven configurations based on the KBS-3 disposal concept, involving variations in geometry, fuel composition (fresh vs. spent), and environmental conditions (e.g., air, argon, flooding scenarios). Effective multiplication factors (k_{eff}) and leakage fractions were evaluated for both codes. Across all cases, OpenMC systematically yielded slightly lower k_{eff} values than MCNP, with absolute code-to-code differences bounded by 136(10) pcm. In the normal-operation configurations, differences were $-32(3)$ pcm (spent) and $-38(3)$ pcm (fresh), with negligible leakage. Under periodic boundary conditions, both codes showed the expected boundary-induced rise in k_{eff} ; for both OpenMC and MCNP, the increase relative to vacuum was $+237(3)$ pcm. By providing reproducible input decks and quantifying agreement under equivalent modeling assumptions, this study represents the first systematic OpenMC–MCNP benchmark for KBS-3-type spent fuel canisters. These findings support extending the validation domain of OpenMC toward spent fuel transport and geological disposal applications.

1. Introduction

The world annual reactor-related uranium requirements are projected to increase from 60,114 tU/year in 2020 to between 63,040 and 108,272 tU/year by 2040 [1]. This increase in uranium demand will lead to a corresponding rise in spent nuclear fuel (SNF) inventories over the coming decades. Addressing the challenges of SNF management, the International Atomic Energy Agency (IAEA) has highlighted the critical need for developing public acceptance, long-term funding, and disposal facilities for high-level waste (HLW) and SNF when considered as waste [2]. Given this scenario, research and development on techniques related to SNF management will become increasingly relevant.

After being used in the reactor, nuclear fuel is removed from the reactor core and can be handled following two main types of fuel cycles: (i) the closed fuel cycle, in which spent fuel is considered to provide a potential future energy resource; and (ii) the open fuel cycle, in which spent fuel is considered as waste [2]. In the open fuel cycle option, spent fuel is stored for several decades to allow the decay heat to be reduced. After a period of storage, the spent fuel is encapsulated in

a robust and corrosion-resistant container to meet disposal acceptance criteria and is then disposed in a deep geological repository (DGR) [2].

A disposal method, defined as the intentional emplacement of radioactive waste in a facility without the intent to retrieve [2] (i.e., permanent storage), is widely accepted among technical experts as the preferred method for ensuring the long-term safety of SNF and HLW. The development of deep geological repositories (DGRs) has been the focus of the disposal strategies of many countries [2].

In the 1970s, Swedish power utilities formed the SKB Nuclear Fuel Safety Project, which developed a disposal method compatible with a DGR called KBS (*Kärnbränsle Sakerhet*, Nuclear Fuel Safety in English), where canisters are placed in deposition tunnels at a depth of about 420 m below the ground surface into individual vertical deposition holes, referred to as KBS-3V (V for vertical) [3]. The original design included a canister with a lead basket for storing fuel rods extracted from spent fuel assemblies, encased in a copper overpack [4]. Later, the lead insert was replaced by a cast iron insert, and fuel assemblies are now disposed as a unit [5]. This approach has influenced the designs

* Corresponding address: CEFNEN, Comisión Chilena de Energía Nuclear, Nueva Bilbao 12501, Santiago, Chile.

E-mail address: jaime.romero@cchen.cl (J. Romero-Barrientos).

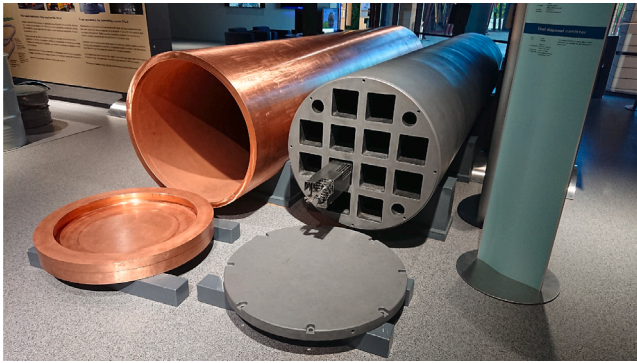


Fig. 1. Olkiluoto final disposal canister (TeemuVäisänen, CCBY-SA4.0). On the left, a copper overpack; on the right, a cast iron insert. In this work, a canister with PWR dimensions was considered for the simulations.

of the Finnish company Posiva Oy [6], which has made significant progress in implementing the KBS-3 method (the modern design is shown in Fig. 1).

A criticality safety evaluation (CSE) is defined as “the analysis and documentation that fissionable material processes remain subcritical under both normal and credible abnormal conditions” [7]. Therefore, the effective multiplication factor (k_{eff}) will be a key parameter in this work, as our main comparison quantity. According to paragraph 6.6 of the International Atomic Energy Agency (IAEA) safety guide SSG-15, “Only verified and validated methods should be used for predicting the safety of operational states or the consequences of accidents” in SNF storage facilities [8]. Thus, in this context, to conduct a CSE and use its results for safety predictions, it is mandatory to use a validated code (i.e. a computational predictive method). Validation and verification of computational methods are typically based on the ICSBEP Handbook, which 2021 version contains a set of 587 criticality-related evaluations [9]. Studies related to SNF storage or disposal methods calculations have been conducted using codes such as MCNP [10], SCALE [11], Serpent [12], TRIPOLI [13], GEANT4 [14] and MCKeff [15]. A study that compares criticality calculations of KENO¹ and OpenMC over decay time using a spent fuel cask has been done in the past with satisfactory results [16], supporting the interest on the application of OpenMC on spent fuel disposal systems. Within the variety of Monte Carlo codes, MCNP, SCALE, Serpent and TRIPOLI are subject to export control [17] regulations. Therefore, there is the need for an accessible Monte Carlo code for this kind of applications and OpenMC appears as a promising option. While numerous benchmarks exist for reactor physics applications, criticality benchmarks specifically addressing KBS-3-type spent fuel canisters are not yet available in the open literature. In particular, no systematic studies have explored the role of boundary conditions in such configurations, nor have direct code-to-code comparisons between MCNP and OpenMC been reported for spent fuel storage systems. The present work does not aim to establish a benchmark of the scope required for inclusion in ICSBEP; rather, it represents an initial step toward addressing this gap by providing reproducible simulations that quantify code-to-code differences in k_{eff} and neutron leakage under representative storage and contingency conditions. Building on this motivation, the present work aims to: (i) develop matched OpenMC/MCNP models of a KBS-3-type canister under normal and contingency-like environments; (ii) quantify code-to-code differences in k_{eff} and leakage; (iii) assess the impact of boundary conditions (vacuum vs. periodic) on lattice-like configurations; and (iv) publish matched input decks to foster reproducibility. The working hypothesis of this study is that OpenMC, when set up with materials,

geometries, boundary conditions, and source definitions equivalent to those in MCNP, can reproduce the corresponding MCNP results for KBS-3-type spent fuel canisters. This hypothesis is tested by comparing k_{eff} and neutron leakage between the two codes under identical modeling assumptions.

2. OpenMC Monte Carlo code

OpenMC is an open source Monte Carlo particle transport code developed at MIT and published in 2015. It is currently maintained and continuously developed by an international community. Originally published with a focus on neutron criticality calculations [18], OpenMC is currently capable of performing fixed source, k-eigenvalue, and subcritical multiplication calculations on models built using either a constructive solid geometry or CAD representation. In addition to these capabilities, OpenMC is able to programming pre- and post-processing, multi-group cross section generation, workflow automation, depletion calculations, multiphysics coupling, and the visualization of geometry and tally results. Nuclear data is handled in HDF5 format; and it can be run in parallel using a hybrid MPI and OpenMP programming model, which has been extensively tested on leadership class supercomputers [19]. Several validation and verification studies have been made, mainly in nuclear reactor systems [20], including a comparison with MCNP and experimental data for time dependent phenomena, obtaining satisfactory results (e.g., a maximum of 0.3% difference between MCNP and OpenMC for the k_{eff}) [21]. Nevertheless, at the moment of writing this article, no public criticality safety evaluations of spent nuclear fuel canisters using OpenMC were found in the literature.

3. Methodology

In this work, criticality calculations were performed for a spent nuclear fuel disposal configuration using OpenMC and MCNP, with the aim of exploring the applicability of OpenMC to this type of system. By benchmarking against MCNP, this study offers supportive evidence for the use of OpenMC in spent fuel canister analyses — a domain where its application has been limited so far. For this study, SNF canisters were simulated in a normal operation situation and in a set of cases that include an external water flood and/or leakage into the canister (described in Section 4.2). Both spent and fresh fuel were considered for all calculations, except for a case where the focus was on the boundary condition of the simulation rather than on the SNF behavior. For each case, the effective multiplication factor was calculated using OpenMC and MCNP. Throughout this study, objects, materials, and environments represent simplified yet representative geometries intended for benchmarking purposes. To validate OpenMC results, the same simulations were performed using MCNP writing equivalent inputs (using the same materials, surfaces and cells definitions), the same nuclear cross section library, and boundary conditions. The initial neutron source was defined as isotropic, located at the same initial position, and with identical numbers of source particles, inactive and active batches. For the energy distribution, a Watt spectrum was adopted in both OpenMC and MCNP, ensuring full consistency between the simulations. These parameters are shown in Table 1. Both MCNP and OpenMC models were built using constructive solid geometry (CSG). The canister internals are represented as a 17×17 PWR fuel-rod lattice using each code’s native lattice capability (MCNP LAT/FILL; OpenMC Lattice). Each pin is modeled with concentric regions — UO₂ pellet, He-filled gap, Zircaloy cladding — and surrounded by the corresponding cover gas/coolant and structural regions (spacer zones and canister shells). All linear dimensions (radii, pitches, lengths, clearances) were taken from the same dimensional tables and applied identically in both inputs, ensuring a one-to-one mapping of surfaces and cells. Global extents and outer boundary conditions (vacuum or periodic) were matched across codes, and the same lattice pitch, array size, and coordinate origin were used to guarantee geometric equivalence. All cases

¹ Part of the SCALE Code System.

Table 1

Input parameters for the simulations executed in this work. Periodic boundary conditions were used only in the lattice test intended to probe potential inter-canister interaction; all other cases used vacuum boundaries.

	OpenMC	MCNP
Version	OpenMC(TD) Ver. 2.1 [21]	MCNP6, 1.0 [23]
Cross sections library	ENDF/B-VIII.0 [24]	ENDF/B-VIII.0 [24]
Particles per batch/hist. per cycle	50,000	50,000
Batches/cycles	3,000	3,000
Inactive batches/cycles	300	300
Initial source position ^a	x,y,z = 0, 0, 1 cm	x,y,z = 0, 0, 1 cm
Initial source angular distribution	Isotropic	Isotropic
Boundary condition(s)	Vacuum (periodic in lattice test)	Vacuum (periodic in lattice test)

^a Given that the coordinate system origin (0,0,0) coincides with a defined surface in the geometry, the particle source was positioned at (0,0,1) cm, applying a 1 cm offset in the z-direction to avoid conflicts with the initial cell assignment during simulation initialization.

were run with fixed statistics (batches/particles per batch and inactive batches held constant across codes). The target statistical precision for k_{eff} was ≤ 10 pcm (one standard deviation). Convergence was assessed by stabilization of the mean k_{eff} after the inactive cycles and by the reduction of the batch-wise standard deviation to the stated target. No adaptive or mesh-based variance reduction was used. Implicit capture (survival biasing) was enabled in OpenMC to mirror MCNP's default behavior, and no additional Russian-roulette/splitting schemes were applied. Thermal scattering data $S(\alpha, \beta)$ for hydrogen in light water (H in H₂O) were consistently applied in both codes wherever water was present. With the execution parameters in Table 1 (50,000 particles per batch; 3000 total batches; 300 inactive), the resulting 1σ statistical uncertainty of k_{eff} in these problems is typically $2\text{--}7 \times 10^{-5}$ (2–7 pcm), as reflected by the uncertainties in parentheses in Tables 3–6. Code versions and nuclear data are also summarized in Table 1. The comparison follows a compact, reproducible protocol: (i) matched materials, geometry, and outer boundary conditions (vacuum or periodic, as applicable), built with CSG and using identical dimensional tables in both inputs; (ii) identical initial source (isotropic, same position) and energy distribution (Watt spectrum), with the same particles per batch, number of batches, and inactive batches (Table 1); (iii) common nuclear data (ENDF/B-VIII.0) and consistent use of $S(\alpha, \beta)$ for H in H₂O wherever water is present; (iv) matched capture/variance treatment (implicit capture/survival biasing enabled in OpenMC to mirror MCNP's default; no additional variance-reduction schemes); (v) metrics and comparisons: the primary metric is the code-to-code difference in k_{eff} reported in pcm; a secondary indicator is the leakage fraction L (fraction of histories escaping the system boundaries in vacuum cases; identically zero under periodic BC). Within-code differences (e.g., fresh vs. spent fuel) and boundary-condition impacts are also reported. Unless otherwise noted, uncertainties are one-sigma statistical errors; uncertainties on differences are obtained by quadratic propagation of the individual k_{eff} uncertainties; and (vi) post-processing and reporting: k_{eff} values and their uncertainties are taken directly from the codes; leakage uncertainties reported as zero are replaced by a minimum of 2 pcm (statistical resolution limit) for consistency (see Section 7). Input decks for a representative case are available at Zenodo [22]. To compare the OpenMC results with the MCNP reference, this work reports the difference in pcm, denoted as “*”Diff. (pcm):

$$\text{Diff. (pcm)} = (k_{\text{OpenMC}} - k_{\text{MCNP}}) \cdot 10^5. \quad (1)$$

Here, k_{OpenMC} and k_{MCNP} are the effective multiplication factors obtained with OpenMC and MCNP, respectively.

4. Systems description

4.1. Canister and fuel assemblies

For the specifications of the SNF canister, geometries and materials were retrieved from a report by Posiva Oy [6], which includes three types of insert (an example is shown in Fig. 1); for this work, the PWR

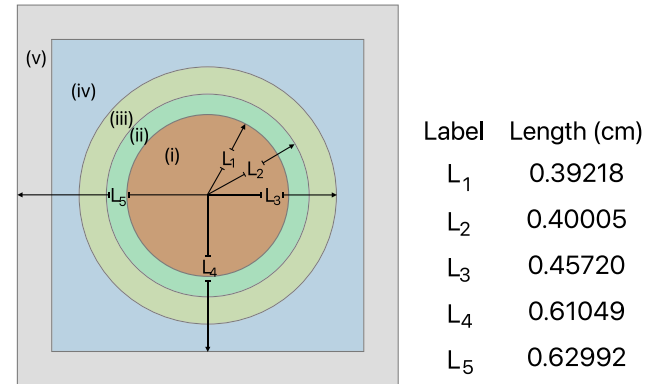


Fig. 2. Radial cross-section of the fuel rod model used in the simulations. The regions labeled (i) to (v) correspond to: (i) fuel, (ii) gas gap, (iii) cladding, (iv) inner spacer boundary, and (v) outer spacer boundary. Distances L_1 through L_5 indicate radial positions from the center to the corresponding region interfaces. The table on the right lists the exact values of each radial dimension, expressed in centimeters. These geometric parameters define the fuel rod structure implemented in the Monte Carlo model. Note that the diagram is not to scale and is intended solely for schematic illustration [26].

type was used since corresponds to the most common kind of reactor intended for energy generation [25].

Regarding the geometric specifications of the fuel assemblies, a simplification of the PWR assemblies described on the BEAVRS benchmark [26] was considered. The radial and axial geometry of the fuel rods and the radial geometry of the spacers were kept. In this study, 17×17 spaces assemblies were considered (with all 289 spaces filled with fuel rods), each rod containing a 380.416 cm long fuel cylinder with the radial geometry shown in Fig. 2. Radially, the fuel rod as a unit consist of a series of layers, the most internal is a solid piece of fuel, followed by a thin layer of fission-produced gas, an M5 cladding that seals the fuel, an intermediate space filled with argon and an M5 frame that sets the border to the neighboring rod.

The nozzles are one of the main parts of the support and alignment structures, they also stay with the fuel rods at the final disposal. In order to emulate the parts that composes the nozzles (i.e. springs, support legs, screws, and others), an homogeneous mixture of 50 v.%² steel and argon was made, similar to the proposal from BEAVRS [26]. Internal spaces in the fuel bundles and gaps within the canister are filled with argon by default in this work, this is corresponding with designs from Posiva and it is due to advantageous thermal and chemical properties from this gas [3,6], exceptions are described in the further subsection.

The OpenMC and MCNP input files used in this study, including geometry, materials, and simulation parameters for the *Normal Operation*

² In this work v.% is the volume proportion.

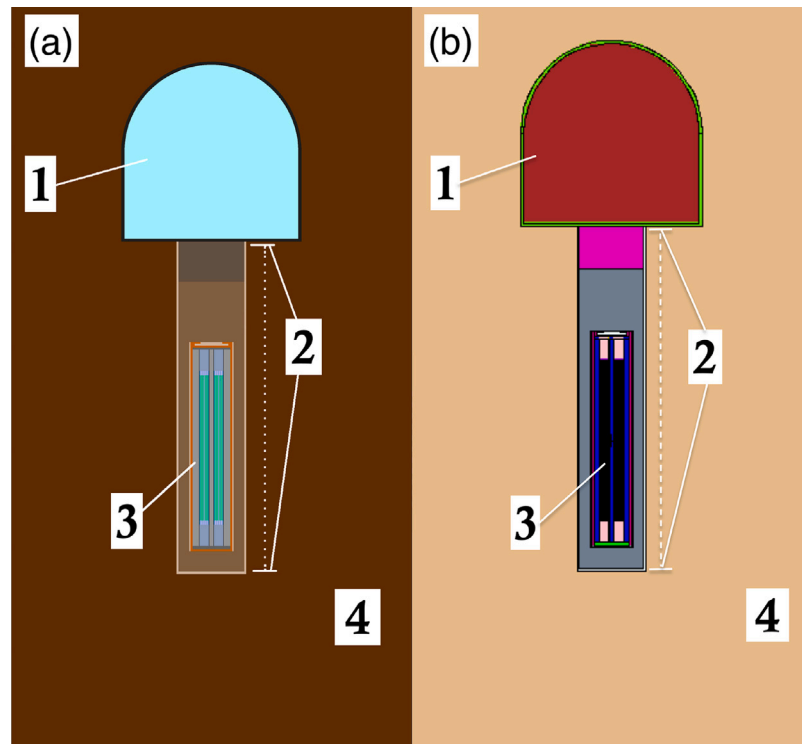


Fig. 3. Normal-operation (NO) environment, YZ view. Panel (a) shows the configuration modeled in OpenMC, and panel (b) the equivalent configuration modeled in MCNP. Numbers 1–4 indicate the same components in both simulations: 1—Tunnel, 2—Borehole, 3—Canister, 4—Surrounding rock.

– *Spent Fuel* configuration, have been made publicly available through Zenodo [22]. This aims to promote transparency and reproducibility of the benchmark results.

In this work, two canister environment sets were considered for the canister location: *Normal operation* and *Flood*; each of them is detailed in the next subsection.

4.2. Canister environments and specifications

In criticality safety studies, boundary conditions are often selected to yield conservative estimates of the multiplication factor, with periodic or reflective surfaces commonly used to overestimate k_{eff} . In this work, vacuum boundary conditions were applied to the *Normal Operation* (NO) and *Flood* (FLD) scenarios, except in one case where periodic boundaries were tested. While vacuum conditions may underestimate k_{eff} due to particle loss at boundaries, this effect is expected to be minimal in large systems like the NO environment, where most particles are absorbed before reaching the system boundary. This consideration is relevant for interpreting potential differences between codes, particularly in configurations where neutron leakage is non-negligible.

In OpenMC, the leakage fraction provides a measure of how many particles escape the geometry without undergoing further interactions. A near-zero leakage value indicates that particles are predominantly terminated by nuclear reactions (e.g., absorption), while higher leakage suggests potential underestimation of the multiplication factor due to boundary losses. For the NO environment, which features large boundaries, leakage is expected to be negligible. In contrast, the FLD cases — which represent more compact or hypothetical geometries — can exhibit non-negligible leakage. To assess its impact, an additional comparison between vacuum and periodic boundary conditions was performed.

4.2.1. Normal operation environment

Criticality calculations for these canisters made by Posiva locates the canister in a fitted box (considering that the canister is 1.05 m wide),

that either will have a $1.1 \times 1.1 \text{ m}^2$ base for an isolated dry canister and periodic lattice array; or a $1.6 \times 1.6 \text{ m}^2$ base for an isolated flooded canister [27]. On the one hand, simple and fit environments similar to the ones considered in this work are common in criticality evaluations for SNF canisters [28–30]. On the other hand, wider environments have been used in other studies, considering simplified materials and structures, with sizes up to an estimated $6 \times 6 \times 10 \text{ m}^3$ box [31–33].

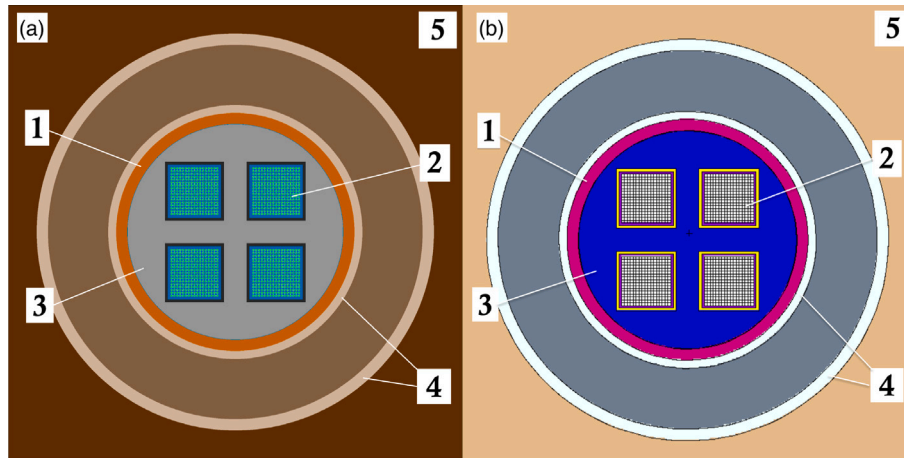
The NO environment considers a deep geological repository scenario, which consists on a concrete coated tunnel filled with air and a vertical hole to locate the canister filled with bentonite and other construction materials [31]. Outside these structures there is an upper crust rock filling. The geometry for this environment is based on thermal analyses that contain more detail on the structures surrounding the canister [3,31], obtaining a high detail $10 \times 10 \times 20 \text{ m}^3$ environment shown in Figs. 3 and 4(a).

4.2.2. Flood

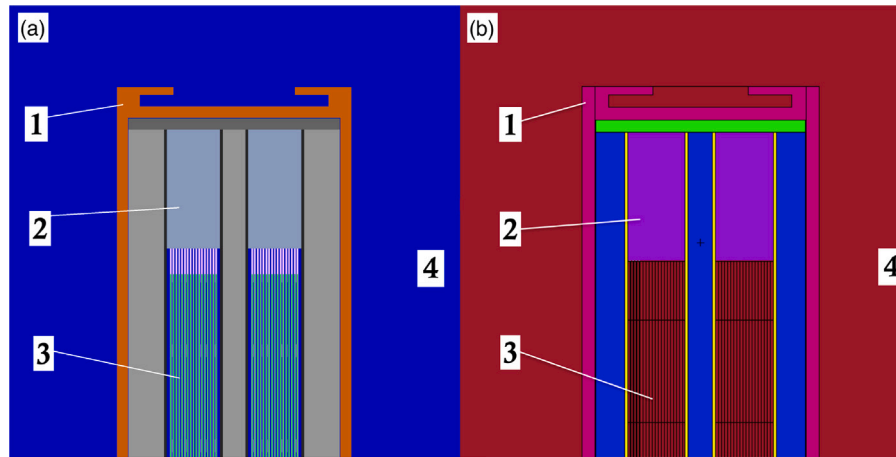
In the FLD environments, the canister is located in a $5 \times 5 \times 10 \text{ m}^3$ box that could be either filled with air or water. The insides of the canister can be either unchanged (argon) or emulating a water leakage, in the latter, the argon gas in the empty spaces inside the canister has been replaced by water and the nozzle material is now a mixture of steel and water. The presence of water through either an external flood or a leakage are considered as contingencies since the increase of hydrogenated materials near the SNF provokes an increase in neutron moderation through elastic scattering, and, therefore, an increase in the k_{eff} . These variations give rise to four distinct configurations, which in this study are classified based on the material inside the canister (*in*) and the material filling the surrounding environment (*out*). Fig. 4(b) illustrates the Water in – Water out geometry.

4.2.3. Boundary condition change

In addition to the previously described environmental configurations, this study also explored the impact of boundary conditions. When a particle reaches the external limits of the simulation geometry,



(a) Normal-operation (NO) environment, XZ view, zoomed in. Panel (a) shows the configuration modeled in OpenMC, and panel (b) the equivalent configuration modeled in MCNP. Numbers 1–5 indicate the same components in both simulations: 1: Copper overpack. 2: Fuel assembly. 3: Iron insert. 4: Bentonite pellet layer. 5: Rock surrounding.



(b) Flood (FLD) environment, "Water-in/water-out" case, YZ view, zoomed in. Panel (a) shows the configuration modeled in OpenMC, and panel (b) the equivalent configuration modeled in MCNP. Numbers 1–4 indicate the same components in both simulations: 1: Copper overpack. 2: Assembly nozzles. 3: Water surrounded fuel rods. 4: Water surrounding.

Fig. 4. Visualization of the main parts of the simulated systems. Made with the OpenMC plot tool.

a *vacuum* boundary condition terminates its track and simulation. A *periodic* boundary condition, on the other hand, transports the particle from the border surface to a *pair* surface set by the user (usually at the opposite side of the system), keeping the particle within the geometry and allowing the continuation of its simulation. This algorithm effectively emulates the repetition of the system an indefinite number of times. Several canisters placed in proximity may exhibit neutronic coupling due to inter-canister neutron transport. To examine how such coupling might influence simulation results, the *Argon in – Air out* case with spent fuel was modeled using both vacuum and periodic boundary conditions. The difference in k_{eff} values between these two boundary types provides a reference for assessing the sensitivity of the system to inter-canister interactions. This scenario was chosen as a simplified dry storage configuration, and serves to explore how boundary conditions may affect k_{eff} in both OpenMC and MCNP.

4.3. Materials

Initially, the SNF isotopic specification was obtained from a benchmark PWR SNF composition intended as a starting point for decay

calculations [34], this composition contains 112 nuclides with atomic weight fractions ranging from 2.73×10^{-21} to 6.72×10^{-1} .

For this work 1.5×10^8 particles were simulated for each case studied, implying that, in the case that every particle interacts with the SNF, a minimum concentration of $\sim 0.67 \times 10^{-8}$ is necessary for a nuclide to have an average of interactions greater than 1 per simulation. Therefore, it can be considered that nuclides with a concentration of 10^{-9} or less will not be object of interaction.

Since the main objective of this work is evaluating the OpenMC application on SNF criticality safety evaluations through a benchmark with MCNP, a set of justified simplifications was applied to the SNF composition, consistent with the goal of benchmarking OpenMC against MCNP in representative scenarios. As a simplification measure, three steps were applied: (i) nuclide concentrations were reduced by decay over a period of 30.5 days (~ 1 month), based on their half-lives [24]; this duration was selected to represent a plausible timeframe for SNF transport and logistics, while omitting extended decay pool storage for modeling simplicity; (ii) only nuclides with a concentration fraction above 10^{-9} were retained, resulting in a set of 60 nuclides—among which three (^{14}C , $^{129\text{m}}\text{Te}$ and $^{242\text{m}}\text{Am}$) are not included in the ENDF/B-VIII.0 cross section library; and (iii) these three nuclides were excluded

Table 2
Materials considered in the simulations.

Part(s)	Material	Specifications and/or reference
Spent Fuel	Spent UO ₂ based fuel	initial 4.5 wt% ²³⁵ U, 50 GWd/MTU [34]
Fresh fuel	Uranium dioxide	4.5 wt% ²³⁵ U
Spent Inner rod gas	Helium	[26]
Fresh Inner rod gas	Fission gas	[35]
Fuel rod cladding & spacers	M5	with maximum impurity concentration [36]
Assembly empty space filling	Argon	natural abundance composition, 1 atm [6,37]
Nozzle	Steel & argon	S355JR EN10025-2:2004, 50 v.% each [38]
Profile	Steel	EN 10210 S355J2H [38]
Insert	Cast iron	EN(−1563)-GJS-400-15(GGG40) [39]
Insert lids	Steel	S355JR EN10025-2:2004 [40]
Overpack	Oxygen free copper	C102 standard [41]
Filling pellet	Bentonite & air	MX80 bentonite, 50 v.% air [37,42]
Hole buffer	Bentonite	MX80 bentonite [42]
Tunnel coating	Projected concrete	steel-reinforced [38,43–45]
Dense backfill	Bentonite & cement	50 v.% each [42,44]
Tunnel filling	Air	1.327 wt% water [46]
Outside of the tunnel	Continental crust rock	[47]
Flooding environment	Pure light water	[37]

from the simulations, yielding a final spent fuel composition of 57 nuclides. To provide a reference for comparison, fresh fuel was modeled as UO₂ with an initial ²³⁵U enrichment of 4.5%, consistent with typical PWR specifications. The spent-fuel inventory resulting from the three-step screening above (57 nuclides) was used *without* depletion; all operations (30.5-day decay, thresholding, and library check) were carried out as an external pre-processing step. The resulting atomic fractions were then entered verbatim in both MCNP and OpenMC at the same material densities. The complete nuclide list and number densities used in the simulations are provided in the openly available input decks at Zenodo [22], ensuring transparency and reproducibility. For the composition of other materials, certain standards were considered -based on technical reports made by Posiva Oy [3,6]- which can be found in the third column of Table 2.

5. Results and discussion

In this section, the obtained effective multiplication factors will be presented and compared following Section 3. A comparison between results for both codes was performed on progressively more complex systems, from a fuel rod section to the full canister, using fresh fuel and argon canister filling on every system.

5.1. Normal operation

In this environment, code-to-code differences were −32(3) pcm for spent fuel and −38(3) pcm for fresh fuel, as shown in Table 3. Replacing spent fuel with fresh fuel increased k_{eff} by 8939(3) pcm in OpenMC and by 8945(3) pcm in MCNP. This increase in reactivity is expected from the higher content of fissile ²³⁵U in the fresh fuel. Furthermore, in the spent fuel a fraction of the ²³⁵U inventory has been consumed and replaced with a wide set of fission products and minor actinides, some of which act as neutron absorbers, thereby further reducing the system reactivity.

5.2. Flood

Flooded (FLD) scenarios were evaluated for four medium combinations inside/outside the canister: water-in/water-out, water-in/air-out, argon-in/water-out, and argon-in/air-out, for both spent fuel (SF) and fresh fuel (FF). For SF (Table 4), the cases with water-in (water-in/water-out and water-in/air-out) yield $k_{\text{eff}} \approx 0.78$, with OpenMC–MCNP differences of −108(9) and −86(8) pcm, respectively. When the canister interior remains argon (argon-in/water-out and argon-in/air-out), the system is deeply subcritical ($k_{\text{eff}} \approx 0.18$ –0.184), and the code-to-code differences drop to the tens-of-pcm range (e.g., −21(7)

Table 3

Comparison of k_{eff} in the NO environment for spent and fresh fuel. *Diff. (pcm)* is defined as $(k_{\text{OpenMC}} - k_{\text{MCNP}}) \times 10^5$; the quoted uncertainty on *Diff. (pcm)* is the one-sigma value obtained by quadratic propagation of the individual k_{eff} uncertainties. *Leakage* is the fraction of histories escaping the model boundary. [*]: Leakage uncertainty reported as zero by OpenMC was replaced with a minimum value of 2 pcm (statistical resolution limit, 1/50,000) to avoid misleading precision. See Section 7 for details.

Case	k_{eff} OpenMC	k_{eff} MCNP	Diff. (pcm)	Leakage
Spent Fuel	0.19556(2)	0.19588(2)	−32(3)	0.00001(2 *)
Fresh Fuel	0.28495(2)	0.28533(2)	−38(3)	0.00001(2 *)

Table 4

Comparison of k_{eff} in flooded (FLD) scenarios for spent fuel. *Diff. (pcm)* is defined as $(k_{\text{OpenMC}} - k_{\text{MCNP}}) \times 10^5$; the quoted uncertainty on *Diff. (pcm)* is the 1 σ value obtained by quadratic propagation of the individual k_{eff} uncertainties. *Leakage* is the fraction of histories escaping the model boundary. [*]: Leakage uncertainty reported as zero by OpenMC was replaced with a minimum value of 2 pcm (statistical resolution limit, 1/50,000) to avoid misleading precision. See Section 7 for details.

Case	k_{eff} OpenMC	k_{eff} MCNP	Diff. (pcm)	Leakage
Water-in/water-out	0.78131(7)	0.78239(6)	−108(9)	0.00000(2 *)
Water-in/air-out	0.78051(6)	0.78137(6)	−86(8)	0.07844(2)
Argon-in/water-out	0.18439(7)	0.18460(2)	−21(7)	0.00000(2 *)
Argon-in/air-out	0.18189(2)	0.18218(2)	−29(3)	0.49921(4)

to −29(3) pcm). As expected, leakage is negligible when the exterior medium is water and becomes appreciable with air outside: about 0.078 for water-in/air-out and about 0.499 for argon-in/air-out. For FF (Table 5), the water-in cases become supercritical ($k_{\text{eff}} > 1$). Despite the stronger moderation, the OpenMC–MCNP differences remain small, with a maximum absolute difference of 136(10) pcm (water-in/water-out). Leakage follows the same pattern: effectively zero with water outside and non-negligible with air outside. Overall, the trends are physically consistent across both codes: moderation by water increases k_{eff} , air outside increases leakage and slightly depresses k_{eff} , and OpenMC systematically reports slightly lower values than MCNP by at most a few hundred pcm in the most reactive FLD cases.

5.3. Boundary type change

To assess the potential for neutronic coupling between neighboring SNF canisters, a comparison was performed between vacuum and periodic boundary conditions for the configuration with the highest observed leakage (argon-in/air-out). Switching to periodic boundaries increased k_{eff} by +237(3) pcm in both OpenMC (from 0.18189(2) to

Table 5

Comparison of k_{eff} in flooded (FLD) scenarios for fresh fuel. *Diff. (pcm)* is defined as $(k_{\text{OpenMC}} - k_{\text{MCNP}}) \times 10^5$; the quoted uncertainty on *Diff. (pcm)* is the 1σ value obtained by quadratic propagation of the individual k_{eff} uncertainties. *Leakage* is the fraction of histories escaping the model boundary. The largest absolute code-to-code difference in these cases is 136(10) pcm. [*]: Leakage uncertainty reported as zero by OpenMC was replaced with a minimum value of 2 pcm (statistical resolution limit, 1/50,000) to avoid misleading precision. See Section 7 for details.

Case	k_{eff} OpenMC	k_{eff} MCNP	Diff. (pcm)	Leakage
Water-in/water-out	1.05392(7)	1.05528(7)	−136(10)	0.00000(2 *)
Water-in/air-out	1.05277(7)	1.05408(6)	−131(9)	0.07659(2)
Argon-in/water-out	0.26517(2)	0.26559(2)	−42(3)	0.00000(2 *)
Argon-in/air-out	0.26101(2)	0.26141(2)	−40(3)	0.48751(4)

Table 6

Comparison of k_{eff} for spent fuel in the argon-in/air-out configuration under vacuum vs. periodic boundaries. *Diff. (pcm)* is defined as $(k_{\text{OpenMC}} - k_{\text{MCNP}}) \times 10^5$; the quoted uncertainty on *Diff. (pcm)* is the one-sigma value from quadratic propagation of the individual k_{eff} uncertainties. For context, the within-code boundary-induced change for both OpenMC and MCNP is +237(3) pcm (periodic minus vacuum). The measured leakage for the periodic system was 0.47077(4).

Boundary type	k_{eff} OpenMC	k_{eff} MCNP	Diff. (pcm)
Vacuum	0.18189(2)	0.18218(2)	−29(3)
Periodic	0.18426(2)	0.18455(2)	−30(3)

0.18426(2)) and MCNP (from 0.18218(2) to 0.18455(2)), as it can be seen in Table 6). These results indicate that canisters placed 3.95 m apart in a simplified dry-storage-like geometry may exhibit inter-canister neutron coupling. Although this scenario is not a full physical model of dry storage, the findings illustrate how boundary treatment and geometric assumptions can influence criticality estimates across simulation tools.

6. Limitations

This work presents a first step toward benchmarking OpenMC for spent nuclear fuel storage applications, and its scope is intentionally limited. The study relies on matched OpenMC/MCNP models and does not include validation against experimental benchmarks, which would be required for regulatory or licensing analyses. The fuel compositions considered represent a specific spent fuel isotopic inventory and a single fresh fuel enrichment, without exploring a broader parameter space. In addition, the analysis focused on k_{eff} and neutron leakage as primary metrics, leaving out spectral comparisons, sensitivity analyses, and uncertainty quantification. These simplifications were adopted to ensure model consistency across both codes and to keep the work within a reproducible framework. Therefore, the findings should be understood as an initial contribution toward extending OpenMC validation, providing a reproducible baseline for future studies rather than a comprehensive criticality safety assessment.

7. Summary & conclusions

In this work, criticality calculations of systems containing KBS-3 spent nuclear fuel canisters were conducted, considering normal operation conditions, a set of abnormal conditions that associate two contingencies, an external flood and a water leakage into the canister, and a periodic lattice boundary condition to briefly explore potential canister-to-canister neutron interaction. Although these abnormal conditions are commonly proposed on Criticality Safety Evaluations, it should be noted that this study is not intended as a regulatory or licensing-oriented safety analysis. The present work focuses solely on benchmarking OpenMC against MCNP by comparing k_{eff} values under equivalent modeling conditions. For regulatory applications, additional steps would be required, including validation against experimental

benchmarks and the integration of safety margins and conservatism, which remain beyond the scope of this paper. The variety of environments are complemented by the replacement of the SNF for fresh fuel for all the mentioned cases. Through the integrity of this study, a PWR waste management case was considered, including the geometry and composition of the SNF, fuel bundles and canister. Considering MCNP as a Monte Carlo reference for OpenMC, a total of 11 benchmark calculations were conducted with equivalent inputs, using the same definitions of materials, cells, and surfaces, the same execution settings, and the same cross-section library. Differences are reported in pcm (Eq. (1)); the quoted uncertainty for “*”Diff. (pcm) corresponds to the propagated statistical uncertainty from the individual k_{eff} estimates. In the normal-operation (NO) environment (Table 3), the code-to-code differences were −32(3) pcm for spent fuel and −38(3) pcm for fresh fuel, with negligible leakage fractions [0.00001(2 *)], as expected for a large, open geometry. While the configurations are simplified, they reflect representative conditions and provide useful benchmarks for assessing OpenMC’s behavior in realistic SNF storage environments. Replacing spent fuel with fresh fuel led to higher k_{eff} in both codes, consistent with the larger ^{235}U content in fresh fuel. For the FLD cases (Table 4 for spent fuel; Table 5 for fresh fuel), the code-to-code differences ranged from 29(3) pcm (argon-in/air-out, spent) up to 136(10) pcm (water-in/water-out, fresh). Systems with water leakage into the canister became supercritical with fresh fuel. In all cases with water-filled boxes, the leakage fraction was null within statistical bounds, whereas air-out boxes exhibited substantial leakage, as expected from the open boundary. To assess potential inter-canister coupling, a comparison between vacuum and periodic boundary conditions was performed for the configuration with the highest leakage (argon-in/air-out, spent). Under periodic boundaries, k_{eff} increased by +237(3) pcm in both OpenMC and MCNP, indicating similar sensitivity in both codes to boundary-induced interaction effects. Although this setting is not a full physical model of dry storage, it illustrates how boundary treatment and geometric assumptions can influence criticality estimates across tools. In several cases, OpenMC reported an uncertainty of zero for the leakage fraction, which is not physically meaningful given the stochastic nature of Monte Carlo simulations. To address this, a minimum uncertainty of 2 pcm was adopted. This value corresponds to the smallest detectable fraction (1/50,000) with 50,000 particles per batch. This choice ensures a conservative and consistent treatment of leakage, avoiding misleading precision and enabling meaningful comparisons across cases and codes. The safety of a spent fuel storage facility — and of the spent fuel stored within it — is ensured by maintaining subcriticality, among other functions [8]. KBS-3 is one of the principal concepts for long-term SNF disposal [2], and prior criticality studies support its effectiveness [27]. Validation of accessible predictive methods should be promoted, particularly in countries without legal access to mainstream Monte Carlo codes validated through standards like the ICSBEP Handbook. Criticality calculations for systems with spent nuclear fuel contribute to expanding OpenMC’s application space, while MCNP–OpenMC benchmarking supplies robust code-to-code validation. Across all benchmark cases, code-to-code differences remained within 136(10) pcm. OpenMC yielded slightly lower k_{eff} than MCNP in every case, indicating a small but consistent bias relative to MCNP under the conditions studied. To further expand OpenMC’s applicability to regulated scenarios, future comparisons against experimental benchmarks involving spent-fuel configurations are recommended; this would bridge code-to-code validation with regulatory-oriented criticality safety evaluations. In addition to the benchmarking results presented, this work highlights a methodology with broader applicability to real-world scenarios. Developing fully equivalent OpenMC and MCNP inputs, systematically comparing k_{eff} and leakage, and openly publishing reproducible models provide a framework that can be directly applied in contexts such as dry cask storage design, repository safety evaluations, and regulatory analyses requiring validated predictive tools. While the present study is not intended as a licensing-oriented safety analysis, the demonstrated agreement under representative KBS-3 conditions supports the potential of OpenMC to be reliably employed in safety-related applications where open-source accessibility is essential.

8. Future work

Beyond the scope of the present benchmark, several directions can extend this research. First, a systematic comparison of neutron spectra between OpenMC and MCNP at selected tally positions and, where useful, flux profiles, could provide complementary insights into transport behavior beyond k_{eff} , which is the central metric of this study. This information can also be relevant in contexts such as shielding, dosimetry, or radiological surveys. Second, beyond the statistical errors reported here, sensitivity analyses and non-statistical uncertainty propagation (e.g., nuclear data covariances, material/dimensional tolerances, and thermal scattering model choices) would help quantify the robustness of the observed code-to-code differences. Third, exploring additional canister-to-canister configurations, such as varying inter-canister distances, would improve the understanding of potential interaction effects in repository-like arrangements. Finally, to further strengthen OpenMC's validation domain, future work should include comparisons against experimental benchmarks involving spent nuclear fuel configurations, thus bridging the gap between code-to-code validation and regulatory-oriented criticality safety evaluations.

CRediT authorship contribution statement

J. Ruiz-Pineda: Writing – review & editing, Writing – original draft, Validation, Investigation, Formal analysis. **J. Romero-Barrientos:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Conceptualization. **F. Molina:** Writing – review & editing, Funding acquisition. **M. Zambra:** Writing – review & editing, Funding acquisition. **F. López-Usquiano:** Writing – review & editing.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (developed by OpenAI) to assist with English grammar revision and preparation of ancillary documents such as the README and metadata files for data publication. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.net.2025.103999>.

Data availability

The input files used to reproduce the results presented in this study are openly available in Zenodo at <https://doi.org/10.5281/zenodo.17106980>. These files correspond to the *Normal Operation – Spent Fuel* configuration and include both OpenMC and MCNP input decks.

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